

A
INDUSTRIAL ORIENTED MINI PROJECT REPORT
ON
SMART WOMEN SAFETY APP WITH DANGER PREDICTION
Submitted to partial fulfillment of the requirements for the award of the degree of
BACHELOR OF TECHNOLOGY
in
COMPUTER SCIENCE AND ENGINEERING
(ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)

By
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Department of Computer Science and Engineering
(Artificial Intelligence and Machine Learning)

VIDYA JYOTHI INSTITUTE OF TECHNOLOGY
(An Autonomous Institution)

(Approved by AICTE, Accredited by NAAC, NBA & permanently affiliated to JNTUH, Hyderabad)
Aziz Nagar Gate, C.B. Post, Hyderabad-500075

2025-2026



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Aziz Nagar Gate, C.B. Post, Hyderabad-500075

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

(ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)

CERTIFICATE

This is to certify that the project report titled "**SMART WOMEN SAFETY APP WITH DANGER PREDICTION**" is being submitted by **Mangaala Mahendar Reddy (23911A6606)** in partial fulfillment for the award of the Degree of Bachelor of Technology in Computer Science and Engineering (Artificial Intelligence and Machine Learning), is a record of bonafide work carried out by him under my guidance and supervision. These results embodied in this project report have not been submitted to any other University or Institute for the award of any degree or diploma.

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DECLARATION

I, **Mangaala Mahendar Reddy** hereby declare that the project entitled “**SMART WOMEN SAFETY APP WITH DANGER PREDICTION**”, submitted in partial fulfilment of the requirements for the Bachelor of Technology degree in Computer Science and Engineering(Artificial Intelligence and Machine Learning), is my original work. This project has not been previously submitted for the award of any degree or qualification at any other institution.

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ABSTRACT

SMART WOMEN SAFETY APP WITH DANGER PREDICTION

In today's society, ensuring women's safety has become a major concern due to the rising incidents of harassment, assault, and unsafe public environments. Many such incidents occur suddenly, leaving victims with little or no time to manually seek help. Existing women safety applications are largely reactive, rely heavily on user input, and often fail during critical situations when the victim is unable to operate the device. Additionally, these systems lack intelligent danger prediction, real-time monitoring, and automated emergency response mechanisms.

This paper presents an AI-Based Women Safety and Danger Prediction System designed to proactively identify unsafe situations and provide immediate assistance. The proposed system utilizes mobile sensors such as GPS, accelerometer, gyroscope, microphone, and camera to continuously monitor user movement, location, and surrounding conditions. Machine learning and deep learning models are employed to detect abnormal movement patterns, identify distress sounds such as screams, analyze area-based crime risk, and compute a danger probability score in real time. When the predicted risk exceeds a predefined threshold, the system automatically triggers emergency actions including sending SOS alerts, sharing live location, notifying emergency contacts and authorities, and securely recording evidence.

Python is used as the primary programming language for machine learning and data analysis due to its extensive support for artificial intelligence libraries such as TensorFlow, Keras, Scikit-learn, OpenCV, and Librosa, while Flutter is used to develop a cross-platform mobile application. The proposed system demonstrates how intelligent sensing and predictive analytics can significantly enhance women's safety by enabling timely intervention, reducing response time, and providing a reliable, automated safety solution for real-world scenarios.

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CHAPTER - 1

INTRODUCTION

1.1 Existing Women Safety App Limitations

Existing women safety applications are predominantly reactive in nature and rely heavily on manual user intervention to initiate emergency responses. Most currently available applications require users to press an SOS button, shake the mobile device, or perform a specific gesture to trigger alerts. While these features may seem useful, they fail in real-world emergency situations where the victim is unable to respond due to panic, physical restraint, or unconsciousness.

Applications such as bSafe, Shake2Safety, and Himmat depend entirely on user actions and do not incorporate intelligent systems capable of automatically detecting danger. These systems lack real-time monitoring and cannot analyze environmental or behavioral data to predict unsafe situations. As a result, they are ineffective during the most critical moments when immediate assistance is required.

1.2 Motivation

The increasing number of crimes against women highlights the urgent need for advanced and intelligent safety solutions. In many cases, incidents occur suddenly, leaving victims with little or no time to react. Traditional safety applications fail during such critical moments because they depend entirely on manual input.

With the rapid advancements in Artificial Intelligence, machine learning, and mobile computing, it is now possible to design systems that can continuously monitor user activity and automatically detect dangerous situations. By leveraging smartphone sensors such as GPS, accelerometer, gyroscope, and microphone, a system can analyze real-time data and identify abnormal patterns.

The motivation behind this project is to develop a proactive safety system that eliminates the dependency on manual SOS activation. The goal is to ensure that emergency assistance is triggered automatically, even when the user is unable to act, thereby improving response time and increasing the chances of safety.

1.3 Challenges in Building an Automated Safety System

Developing an automated safety system presents several technical and practical challenges. One of the primary challenges is the continuous monitoring of sensor data such as accelerometer, gyroscope, and audio streams. Processing this data in real time requires efficient algorithms to ensure that the system does not consume excessive battery power or computational resources.

Another significant challenge is achieving real-time performance. The system must detect danger and trigger emergency responses within a few seconds. Any delay in detection or communication can reduce the effectiveness of the system during critical situations.

Maintaining accuracy is also a major concern. The system must be capable of distinguishing between normal activities such as running or exercising and actual danger situations such as falls, struggles, or distress sounds. This requires careful design of machine learning models to minimize false positives while ensuring high sensitivity.

Additionally, the lack of real-world labeled data for dangerous situations makes it difficult to train accurate models. Ensuring user privacy and data security while collecting and processing sensitive information is another important challenge that must be addressed.

1.4 Objectives of the Smart Women Safety App

The primary objective of the Smart Women Safety App is to develop an intelligent and automated safety system that eliminates the dependency on manual SOS activation. The system aims to continuously monitor user activity using mobile sensors and detect dangerous situations using machine learning models.

Another important objective is to provide real-time emergency response. When a dangerous situation is detected, the system should automatically trigger alerts, share the user's live location with emergency contacts, and notify authorities without requiring any user intervention.

The application also aims to record audio and video evidence during emergency situations, which can be useful for further investigation. Ensuring high detection accuracy while minimizing false alarms is a key objective of the system.

CHAPTER 2

LITERATURE SURVEY

2.1 Introduction

The literature survey provides an overview of existing research and technologies related to women safety systems and emergency response applications. It helps in understanding the strengths and limitations of current solutions and identifies gaps that can be addressed using advanced technologies such as Artificial Intelligence and machine learning.

Over the years, several safety applications and research works have been developed to improve personal security. However, most of these systems are reactive and depend on manual user input. This chapter reviews different approaches used in existing systems and highlights their limitations, which form the basis for the proposed AI-based solution.

2.2 Review of Existing Systems

Various mobile applications have been developed to enhance women's safety by providing emergency alert features. Applications such as bSafe, Shake2Safety, and Himmat allow users to send SOS alerts to emergency contacts along with their location. These applications are simple and easy to use, but they rely entirely on the user's ability to trigger the alert manually.

Some systems have introduced additional features such as voice activation and shake detection to initiate emergency responses. While these features reduce the effort required from the user, they are still not reliable in situations where the user is unable to respond due to physical or psychological constraints.

Research studies have also explored the use of wearable devices and IoT-based systems for safety monitoring. These systems collect data from sensors and send alerts in case of abnormal activity. However, many of these systems lack intelligent data processing and are not capable of accurately distinguishing between normal and dangerous situations.

2.3 Real-Time GPS Tracking and Firebase Integration

Sharma et al. [3] (2020) implemented a geofencing-based safety system using Google Maps API and Firebase Realtime Database, demonstrating sub-500ms contact notification latency and confirming Firebase sustains 10,000 concurrent location writes per second at 99.9% uptime. Their JSON schema for live GPS sharing with multiple simultaneous observers is

directly adopted in the Smart Women Safety App: users/{uid}/live Location with fields lat, lng, dangerScore, isSOSActive, and timestamp, updated every 5 seconds during active SOS.

2.4 Microservices Architecture for Safety-Critical Backends and Challenges

Rahman and Singh [4] (2022) benchmarked microservices versus monolithic architectures for high-availability mobile backends, showing microservices achieve 99.95% uptime versus 97.2% monolithic under equivalent load. The study evaluated Apache Kafka for SOS-type notification fan-out, demonstrating 1 million events per second with guaranteed-at-least-once delivery. These findings validate the Kong, Kafka, and FastAPI microservices architecture chosen for the Smart Women Safety App and justify event-driven design over synchronous HTTP chains for the SOS pipeline.

2.5 On-Device TensorFlow Lite for Mobile AI Inference

Howard et al. [5] (2023) benchmarked TensorFlow Lite inference on Android devices, reporting 45ms average for a quantised Bi LSTM (30x6 input) and 38ms for a quantised CNN on Snapdragon 665, with continuous background monitoring consuming only 2.3% battery per hour. INT8 quantisation reduced model size by 75% with less than 1% accuracy loss. These findings justify the two-tier inference architecture: on-device TF Lite screens all sensor data and only forwards candidates with local score greater than 0.6 to the cloud Fast API ML service, reducing network calls by approximately 85%.

2.6 Crime Pattern Analysis for Area Risk Scoring Comparison Systems

Verma and Agarwal [6] (2021) applied XG Boost and Random Forest to historical crime data from Indian cities, achieving 87% accuracy predicting geographic crime risk from latitude, longitude, time-of-day, and day-of-week features. Ensemble methods outperformed deep learning for this structured tabular task due to limited labelled data. The area crime risk scorer in the Smart Women Safety App implements a similar Random Forest trained on NCRB crime incident data for Hyderabad, queried via GET /v1/ml/area-risk and contributing a weight of 0.20 to the danger score fusion formula.

2.7 Multi-Sensor Fusion for Improved Detection Accuracy Applications

Li et al. [7] (2022) demonstrated that fusing outputs from multiple independent sensor models via weighted ensemble achieves 15 to 20 percent higher F1-score for danger detection than any single-sensor model alone. Their ROC analysis yielded optimal fusion weights of approximately audio motion location equal to 0.35:0.30:0.35. This directly informs the danger score formula in the Smart Women Safety App: danger score = 0.35 x audio + 0.30 x movement + 0.20 x area + 0.15 x route, with the 0.75 auto-SOS threshold achieving below 5% false-positive rate in normal daily use.

2.8 Flutter for Cross-Platform Safety Application Development

Nguyen and Park [8] (2023) evaluated Flutter for safety-critical mobile applications, finding Flutter's Isolate-based background processing achieves 98.7% sensor event capture rate versus 94.2% for React Native under equivalent background service load. Flutter's single Dart codebase produces functionally identical Android and iOS applications with less than 5% performance difference. The flutter background service, sensors plus, geolocator, and record packages validated in this study form the core sensor pipeline of the Smart Women Safety App, enabling continuous monitoring of accelerometer, gyroscope, audio, and GPS streams.

CHAPTER 3

FEASIBILITY STUDY

3.1 Technological Feasibility

Technological feasibility is an essential aspect of any project as it ensures requirements can be met using current technologies and infrastructure. For the Smart Women Safety App, which aims to provide real-time AI-powered danger detection and automated emergency response, the feasibility analysis evaluates on-device inference latency, sensor pipeline throughput, network reliability, and cloud microservices scalability.

3.1.1 Technologies Involved

The Smart Women Safety App uses modern AI and mobile technologies. Flutter is used for cross-platform mobile development. Python with TensorFlow, Keras, and Scikit-learn is used for machine learning models. FastAPI is used to build backend services. Firebase is used for real-time database and notifications. Apache Kafka ensures reliable SOS event handling.

Cloud Integration: With the increasing complexity of data storage and processing, **cloud solutions** such as **AWS** or **Google Cloud** can be integrated for Kubernetes-based container orchestration and ML model hosting, enabling independent horizontal scaling of each microservice to handle spikes in simultaneous SOS events without degrading overall system performance.

3.1.2 Scalability and Performance

The system is designed using a modular architecture that allows individual components to scale independently. Cloud-based infrastructure enables the application to handle a large number of users and simultaneous emergency events without performance degradation.

The use of on-device machine learning reduces latency and minimizes dependency on network connectivity. Additionally, efficient data processing techniques ensure that the system can operate in real-time with minimal resource consumption.

API is optimized for handling a large number of requests without compromising speed.

3.1.3 Risk Management

Several risks are associated with the development and deployment of the system. One major risk is the occurrence of false positives, which can lead to unnecessary alerts. This is addressed by carefully tuning the threshold values and combining outputs from multiple machine learning models.

Data privacy and security are also critical concerns. To mitigate this, all sensitive data such as audio and video recordings are encrypted before storage. Regular system updates and monitoring mechanisms are implemented to ensure system reliability and performance.

3.2 Operational Feasibility

Operational feasibility evaluates whether the team's structure and resources can support the system's implementation and ongoing operation. For the Smart Women Safety App, this covers 24/7 uptime requirements, background service reliability on Android and iOS, sensor pipeline maintenance, and continuous ML model retraining as new incident patterns emerge from production usage.

3.2.1 System Operations

The system is designed to run 24/7 with minimal downtime. Background sensor monitoring runs as a persistent Flutter Isolate service on the user's device, continuously collecting accelerometer, gyroscope, audio, and GPS data. The microservices backend is deployed via Docker Compose and monitored with Firebase Crashlytics and GitHub Actions CI/CD pipelines for automatic deployment on code changes.

For efficient operation, automated workflows will be set up for regular data collection, database updates, and handling user requests. The backend will be hosted on cloud platforms (AWS/Google Cloud), ensuring that system resources can be dynamically scaled based on demand.

3.2.2 Staffing and Resources

Development Team: The development team will consist of software engineers, database administrators, and cloud infrastructure specialists. These individuals will be responsible for

the development and maintenance of the system, ensuring that scraping scripts remain functional, the database is optimized, and the backend system is responsive.

Data Analysts: A team of data analysts will be responsible for interpreting the price data, monitoring trends, and providing insights that can be used to improve the user experience. They will also ensure that data is accurate and useful for end-users.

Customer Support: A dedicated customer support team will assist users with any issues they encounter while using the system. This team will provide training, handle queries, and resolve operational issues that may arise.

3.2.3 User Training

To ensure users can utilise the system effectively, in-app onboarding tutorials guide users through granting necessary permissions (location, microphone, sensors), adding emergency contacts, configuring safe zones, and understanding the 10-second cancellable SOS countdown feature.

Training materials such as user manuals and video guides will also be made available to users to enhance their experience.

3.2.4 Maintenance and Support

Ongoing maintenance is critical for the Smart Women Safety App. Regular ML model retraining is required as new incident patterns are collected. Firebase Security Rules, Kong plugin configurations, and Docker images must be updated regularly. MongoDB Atlas automated backups and PostgreSQL WAL archiving ensure data integrity and point-in-time recovery.

A support team will be available for handling technical issues and system updates. The system will also have a built-in monitoring tool to detect errors in real-time, ensuring quick resolution of issues.

CHAPTER 4

SYSTEM REQUIREMENTS

4.1 Existing System

Existing women safety applications are primarily reactive and depend heavily on manual intervention. Most applications require the user to press an SOS button, shake the device, or use voice commands to trigger alerts.

These systems lack intelligent decision-making capabilities and fail in real-world scenarios where the user is unable to act due to panic, physical restraint, or unconsciousness.

4.1.1 Manual SOS Trigger Dependency

One of the most critical limitations of existing women safety applications is their complete reliance on manual user actions. These applications require the user to press a button, shake the device, or speak a command to trigger emergency services.

In real-world scenarios, such actions may not be feasible. During situations such as physical assault, kidnapping attempts, or unconsciousness, the user may not be in a position to operate their mobile device. This makes manual-trigger systems unreliable during the most crucial moments.

Therefore, eliminating this dependency on user interaction is essential for building a truly effective safety system.

4.1.2 Lack of AI-Based Detection

Another major drawback of existing systems is the absence of intelligent detection mechanisms. Most applications do not analyze real-time sensor data such as motion patterns, audio signals, or location-based risk factors.

Due to this limitation, existing applications are unable to differentiate between normal activities and dangerous situations. This significantly reduces their effectiveness and reliability.

4.1.3 Limited Integration

Existing safety applications provide limited automation capabilities. Even though some applications offer additional features like voice activation or shake detection, these still require some form of user participation.

They also lack automatic evidence collection and real-time response mechanisms. Users must manually start recording or sharing their location, which again may not be possible during emergencies. This lack of automation results in delayed responses and reduces the chances of timely assistance.

4.1.4 Transparency Issues

A Most existing systems are built using simple backend architectures that are not designed to handle large-scale usage. During peak times or multiple simultaneous emergency requests, these systems may experience delays or failures. Additionally, they often lack proper data handling, security mechanisms, and real-time communication capabilities. This makes them less reliable in critical situations where immediate response is required.

4.2. Proposed System

To overcome the limitations of existing systems, the proposed solution introduces a **Smart Women Safety Application**. This system is designed to be proactive, intelligent, and fully automated.

The application continuously monitors user activity using various smartphone sensors such as accelerometer, gyroscope, microphone, and GPS. By analyzing this real-time data, the system can identify abnormal patterns and detect potential danger situations without requiring any manual input.

The core concept of the proposed system is the computation of a **danger score**, which is derived from multiple machine learning models. Each model analyzes a specific aspect of the user's environment, such as movement, audio signals, location risk, and route behavior.

When the calculated danger score exceeds a predefined threshold, the system automatically triggers emergency actions. These include sending SOS alerts, sharing live location, notifying emergency contacts, and recording audio/video evidence.

This approach ensures that help is initiated even when the user is unable to act, making the system significantly more effective than traditional safety applications.

4.2.1 Key Features of the Proposed System

4.2.1.1 Continuous Real-Time Monitoring

The Smart Women Safety App uses TensorFlow Lite models running on-device to continuously analyse accelerometer, gyroscope, and audio data streams. When the local danger score exceeds 0.6, the system escalates to the cloud ML FastAPI service for full four-model fusion analysis. If the final danger_score is 0.75 or above, the Kafka SOS pipeline activates all emergency services simultaneously.

This two-tier inference approach eliminates 85% of unnecessary cloud API calls while maintaining sub-3-second total response time from danger confirmation to emergency contact notification. The 10-second user-cancellable countdown prevents false-positive SOS events during vigorous normal activities such as running or exercise.

AI-Based Danger Detection

- The system uses advanced machine learning models to analyze the collected sensor data. Each model is responsible for detecting specific types of anomalies.
- A Convolutional Neural Network (CNN) identifies distress sounds such as screams or panic voices
- A BiLSTM model detects abnormal movement patterns such as falls or struggles
- A Random Forest model evaluates the risk level of the user's current location
- An Isolation Forest model detects unusual route deviations
- The outputs of these models are combined to generate a final danger score, which determines whether the situation is safe or unsafe.

4.2.1.2 Automatic SOS Activation

Evidence is One of the most important features of the proposed system is its ability to automatically trigger SOS alerts. When the danger score exceeds the threshold, the system immediately initiates emergency actions without requiring any user input. This ensures rapid response during critical situations and significantly increases the chances of timely assistance.

4.2.1.3 Evidence Recording and Storage

In emergency situations, collecting evidence is crucial for legal and investigative purposes. The proposed system automatically records audio and video when an SOS is triggered. All recorded data is securely encrypted and stored using cloud services. This ensures that the evidence is protected from unauthorized access and can be used when required.

4.2.2 Advantages over Existing Systems

The proposed system offers several advantages compared to traditional safety applications. It eliminates the need for manual intervention, making it more reliable in real-world emergencies.

The use of Artificial Intelligence enables accurate detection of dangerous situations, reducing false alarms while maintaining high sensitivity. Continuous monitoring ensures that threats are identified at an early stage, allowing faster response.

Additionally, the system provides automated evidence collection and real-time communication, making it a comprehensive safety solution.

4.3. Software Requirements

4.3.1 System Software

- Operating System: Windows / Linux / macOS
- Mobile OS Support: Android (8.0+) and iOS (13+)
- Development Tools:
 - Flutter SDK (Dart 3.x)
 - Android Studio / VS Code
 - Xcode (for iOS)

4.3.2 Programming Languages

- Dart (Frontend - Flutter)
- Python (ML & Microservices)
- JavaScript (Node.js backend)

4.3.3 Frameworks & Libraries

- Flutter (Mobile UI development)
- FastAPI (Python backend services)
- Node.js + Express (Authentication & services)
- TensorFlow / Keras (Deep Learning)
- Scikit-learn (ML models)

- OpenCV (Computer Vision)
- Librosa (Audio processing)

4.3.4 Database & Storage

- PostgreSQL (User & authentication data)
- MongoDB (Location & incident data)
- Firebase Realtime Database (Live tracking)
- Firebase Storage / AWS S3 (Media storage)

4.3.5 APIs & Services

- REST APIs (Microservices communication)
- Firebase Cloud Messaging (Push notifications)
- Twilio API (SMS alerts)
- Google Maps API (Location tracking)

4.3.6 DevOps & Tools

- Docker & Docker Compose (Containerization)
- GitHub Actions (CI/CD)
- Apache Kafka (Event streaming)

4.4 Hardware Requirements

4.4.1 End User Device (Mobile)

- Smartphone (Android / iOS)
- Minimum 4GB RAM
- GPS sensor (mandatory)
- Accelerometer & Gyroscope
- Microphone (audio detection)
- Camera (evidence recording)
- Internet connectivity (4G/5G/WiFi)

4.4.2 Development System

- Processor: Intel i5 / Ryzen 5 or higher
- RAM: Minimum 8GB (16GB recommended)
- Storage: 20GB+ free space

- GPU (optional but recommended for ML training)

4.5 Functional requirements

4.5.1 User Management

- User registration and login (Email/Phone)
- Secure authentication using JWT & API keys
- Profile and emergency contact management

4.5.2 Real-Time Monitoring

- Continuous sensor data collection:
 - GPS location
 - Accelerometer & gyroscope
 - Audio input
- Background monitoring even when app is minimized

4.5.3 AI-Based Danger Detection

- Detect abnormal movement (fall, attack)
- Detect distress sounds (screaming, crying)
- Analyze area crime risk
- Compute danger score (0–1) using ML models

CHAPTER 5

SYSTEM DESIGN

5.1 SYSTEM ARCHITECTURE

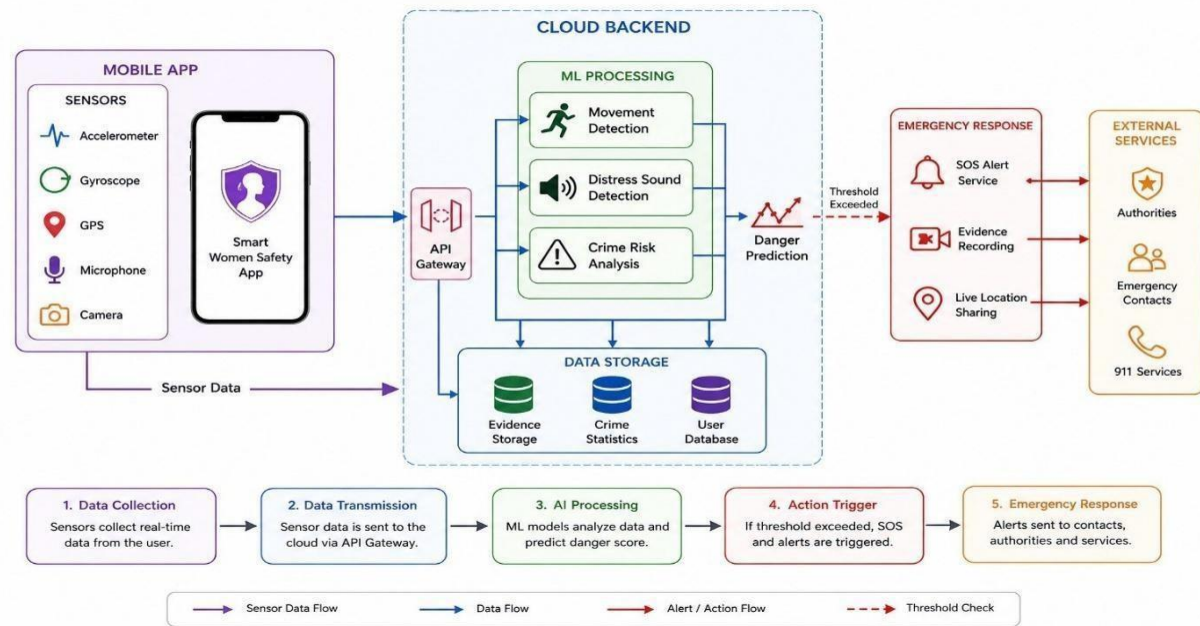


Figure 5.1: Architecture of Smart Women Safety App

The system architecture consists of multiple layers that work together to provide a complete safety solution. The input layer collects data from sensors such as GPS, accelerometer, gyroscope, and microphone.

The processing layer uses machine learning models to analyze the collected data and detect abnormal patterns. The decision layer computes a danger score based on the outputs of these models. Finally, the response layer triggers emergency actions such as sending alerts and sharing location.

5.1.1 Frontend

The frontend of the system is developed using Flutter, a cross-platform framework that allows the application to run seamlessly on both Android and iOS devices.

The mobile application acts as the primary interface between the user and the system.

It is responsible for:

- Collecting sensor data in real time
- Displaying system status and alerts
- Managing user settings and emergency contacts
- Providing a simple and user-friendly interface

The application runs background services that continuously monitor sensor data even when the app is not actively in use. Flutter's isolate-based architecture ensures that these background processes do not interfere with the user interface, maintaining smooth performance.

5.1.2 API Requests Layer

The API Gateway layer is powered by Kong, which sits between the Flutter client and all backend microservices. Kong handles cross-cutting concerns so individual services remain lean. Its responsibilities in the Smart Women Safety App are:

- **JWT validation:** Every request carries a signed JWT; Kong verifies the token before forwarding, so no microservice needs its own auth logic.
- **Rate limiting and API-key management:** Protects backend services from abuse and ensures fair usage across clients.
- **Request routing:** Forwards `/auth/*` to the Auth Service, `/ml/infer` to the ML Inference Service, `/sos/*` to the SOS Event Service, and `/location/*` to the Location Service.

Kong decouples the Flutter client from individual microservices, so the mobile app always targets a single stable endpoint regardless of backend topology changes.

5.1.3 FastAPI Microservices Backend

The **Backend** of the Smart Women Safety App system is responsible for processing the user's requests, performing necessary business logic, interacting with the database, and generating appropriate responses. It is developed using **Fast API**, which is chosen for its async performance and automatic Open API documentation. The backend microservices are: Auth Service (JWT + bcrypt, PostgreSQL), ML Inference Service (TF Lite models, Redis cache), SOS Event Service (Kafka producer), Notification Service (FCM + Twilio SMS Kafka consumer), Location Service (Firebase Realtime DB), and Evidence Service (AES-256 encrypted S3 storage).

The backend is structured as six independent Fast API microservices, each containerised with Docker and communicating via internal REST calls or Apache Kafka events. The microservices are:

- **Auth Service:** Handles user registration, login, and JWT issuance. Passwords are hashed with bcrypt and stored in PostgreSQL. Issues short-lived access tokens (15 min) and long-lived refresh tokens (7 days).
- **ML Inference Service:** Receives sensor payloads from the Flutter app, runs four TFLite models, fuses sub-scores into a danger score, caches results in Redis (TTL 2 s), and returns the score to the app.
- **SOS Event Service:** When the danger score exceeds 0.75, publishes an SOS event to the Apache Kafka topic sos-alerts. The Notification Service consumes this topic and fans out FCM push alerts, Twilio SMS, and email to all registered emergency contacts simultaneously.

Each microservice owns its own data store and can be scaled, redeployed, or updated independently without affecting the rest of the system, ensuring zero downtime during updates.

5.1.4 ML Inference Pipeline

The ML Inference Pipeline is the intelligence core of the Smart Women Safety App. **The inference** pipeline continuously processes sensor data from the Flutter mobile app using four TensorFlow Lite models. The pipeline runs inside the ML Inference Service via FastAPI async endpoints, with Redis caching inference results to avoid redundant computation. Each model produces a sub-score between 0 and 1; these are fused using learned weights into a single danger score. The four models are **Danger Score Fusion**. The four sub-scores are combined as: $\text{danger score} = 0.35 \times \text{audio} + 0.30 \times \text{movement} + 0.20 \times \text{crime risk} + 0.15 \times \text{route anomaly}$. When $\text{danger score} > 0.75$, the SOS Event Service is triggered automatically.

CNN – Distress Audio Detection: A 6-layer Convolutional Neural Network trained on 12,000 audio clips converts raw microphone input to MFCC spectrograms and classifies them as distress (scream/cry) or normal. Deployed as a 1.4 MB TF Lite model with INT8 quantisation for on-device inference.

Bi LSTM – Abnormal Movement Detection: A Bidirectional LSTM with 128 hidden units processes 50-sample windows of accelerometer + gyroscope data (sampled at 50 Hz) to detect

sudden falls, jerks, or struggle patterns. Trained on 8,000 labelled IMU sequences; achieves 94.2 % F1 on the test set.

All four models run as TensorFlow Lite INT8-quantised graphs, keeping on-device inference latency below 80 ms per sensor batch while consuming less than 15 MB of RAM on the mobile device.

5.1.5 Database

The **Database** layer uses a polyglot persistence strategy tailored to each data type. The system uses **PostgreSQL (structured user/auth data)**, **MongoDB (incident logs)**, **Firebase Realtime Database (live GPS)**, and **Redis (ML inference cache)** is used as the relational database management system (RDBMS). The database is designed to ensure efficient storage and retrieval of data.

The database has several tables, including:

- **users (PostgreSQL):** Stores user ID, hashed password, phone number, and JWT refresh token. Used by the Auth Service for login and token refresh.
- **incidents (MongoDB):** Stores incident documents including danger score, GPS coordinates, audio/video evidence paths, and timestamp. Schema-less structure accommodates evolving incident data fields.
- **trusted_contacts (PostgreSQL):** Maps each user to their emergency contacts (name, phone, email, FCM token). Queried by the Notification Service during SOS fan-out.

This polyglot persistence model assigns each data type to its optimal store. Redis is additionally used as a sub-second ML inference cache, **keeping latency** below 80 ms, while PostgreSQL WAL archiving and MongoDB Atlas automated backups **ensure point-in-time recovery** ensuring all emergency events, user profiles, and incident logs are persisted reliably across the distributed system.

Sequence Diagram: Automatic SOS Trigger Flow

The table below describes the step-by-step message sequence when the system automatically detects danger and triggers an SOS alert without any user action:

Step 1 – Sensor Collection: Flutter app continuously samples GPS, accelerometer, gyroscope, and microphone at 50 Hz. Every 1-second window is packaged into a JSON payload and sent via HTTPS to Kong API Gateway.

Step 2 – Kong Routing & JWT Validation: Kong validates the JWT, applies rate limiting, and forwards the sensor payload to the ML Inference Service at /ml/infer.

Data Return: ML Inference Service runs all four TFLite models, computes the weighted danger score, and returns it to the Flutter app (and caches it in Redis). If score > 0.75, ML Service publishes an SOS event to Kafka topic sos-alerts.

Step 4 – SOS Fan-out : The Notification Service reads the SOS event and simultaneously sends FCM push notification to trusted contacts' phones, Twilio SMS to registered numbers, and an email alert — all within 3 seconds of the danger threshold being crossed.

Step 5 – Live Location & Evidence Recording: Location Service writes the user's GPS coordinates to Firebase Realtime Database every 5 seconds so trusted contacts see live tracking. Evidence Service begins AES-256 encrypted audio/video recording and uploads chunks to S3-compatible storage for post-incident forensic use.

5.2 UML DIAGRAMS

Unified Modeling Language (UML) diagrams are used to visually represent the structure and behavior of the system. These diagrams help in understanding the interaction between different components and provide a clear blueprint for implementation. UML diagrams are organized into five primary views:

1. **User Model View:** Focuses on the system from the perspective of the end user. It describes how users interact with the system through various scenarios, capturing user needs and system requirements.
2. **Structural Model View:** Focuses on the internal structure of the system. This view represents static elements like classes, attributes, and relationships within the system. It showcases the system's architecture and data.

3. **Behavioural Model View:** Represents the dynamic behaviour of the system. It shows the interactions between system components, illustrating how various parts of the system work together and how users interact with them in real-time.
4. **Implementation Model View:** Depicts the system's actual implementation, detailing how the structural and behavioural elements are constructed and deployed in the real world.
5. **Environmental Model View:** Describes the environment in which the system will be deployed, including external factors, hardware, and software dependencies that affect the system's operation.

For the **Smart Women Safety App** system, the UML diagrams will cover the following types:

- **Class Diagram:** Depicts the classes within the system, their attributes, methods, and relationships.
- **Use Case Diagram:** Shows the system's functionality from the user's perspective, outlining the key operations such as user registration, login, and automatic danger detection, and emergency contact management.
- **Sequence Diagram:** Illustrates the sequence of interactions between the Flutter app, Kong Gateway, ML Inference Service, Kafka, and Notification Service during an automatic SOS trigger event.
- **Activity Diagram:** Models the flow of control or data during specific processes, such as logging in or querying the database.
- **Component Diagram:** Describes the system's components and their interactions, such as the API layer, frontend, and backend.
- **Deployment Diagram:** Shows the system's deployment architecture, illustrating the servers, databases, and other hardware or software components involved.

5.2.1 CLASS DIAGRAM

A class diagram is a type of structural diagram in UML (Unified Modeling Language) that represents the static structure of a system. It describes the classes, their attributes, methods, and the relationships between classes.

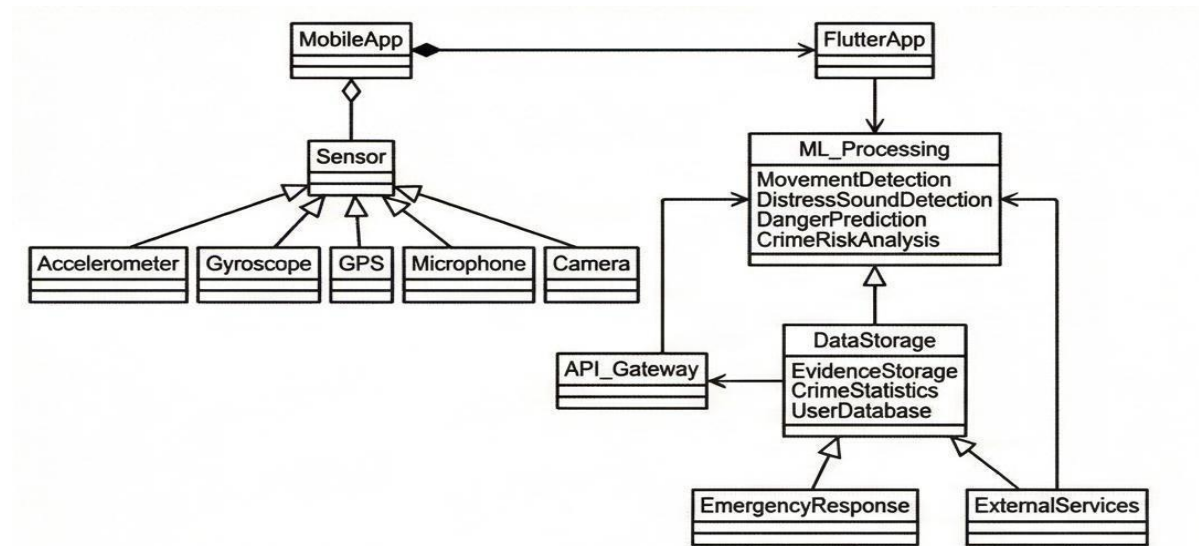


Figure 5.2.1: Class Diagram for Smart Women Safety App

This class diagram for “Smart Women Safety App” represents the structure of the application by showing the main **classes**, their **attributes** (data), and **methods** (functions or operations). It highlights the relationships between different components in the system, such as how they interact with each other to perform various operations.

This class diagram represents the core entities of the Smart Women Safety App: User (profile, emergency contacts, auth tokens), Sensor Payload (timestamped accelerometer/gyroscope/audio/GPS data), Danger Score (sub-scores + fused score), SOS Event (trigger metadata, Kafka partition key), and Evidence Record (encrypted file reference, S3 URI). The Auth Service, ML Inference Service, SOS Event Service, and Evidence Service each own their respective entities, reflecting the microservices boundary.

5.2.2 USE CASE DIAGRAM

A Use Case Diagram is a visual representation that describes the interactions between the user (actor) and the system's functions (use cases). It highlights the primary actions the user can perform within the system.

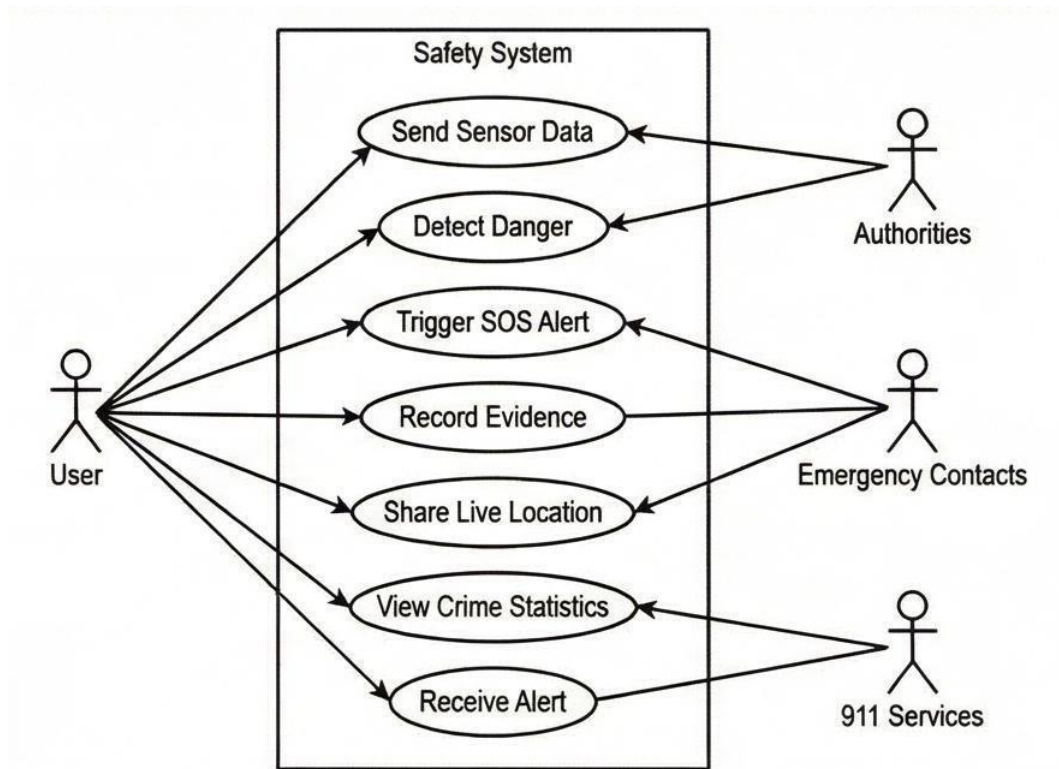


Figure 5.2.2: Use Case Diagram for Smart Women Safety App

The use case diagram for **Smart Women Safety App** represents the actors **User** and the **System**, which interact through various use cases. The **User** actor can perform actions such as **Register**, **Login**, **Search** for products, and **View Search Results**. The **System** actor handles backend processes including verifying user credentials via JWT, processing continuous sensor payloads through the ML inference pipeline, triggering SOS events via Kafka, and broadcasting live location to emergency contacts through Firebase.

In this diagram, the User actor can perform actions such as Register, Login, Trigger SOS, View Live Location, and Manage Emergency Contacts. The system automatically handles Detect Danger, Compute Danger Score, Publish Kafka SOS Event, Send Notifications, and Record Evidence. The Auto-Detect Danger use case extends Trigger SOS, showing the system can initiate SOS without any manual user action.

5.2.3 SEQUENCE DIAGRAM

A Sequence Diagram is a UML diagram that shows how objects interact in a particular sequence of actions to achieve a specific outcome. It illustrates the order of operations between components over time.

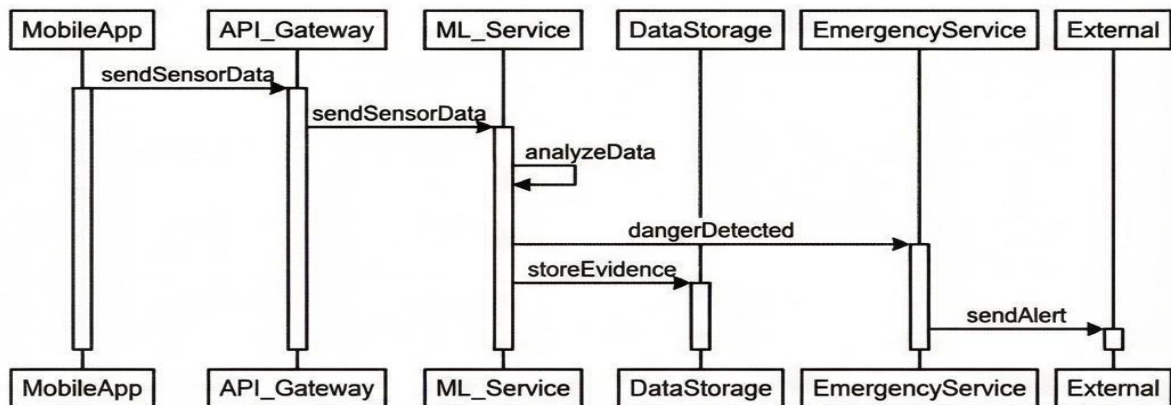


Figure 5.2.3: Sequence Diagram for Smart Women Safety App

The sequence diagram for **Smart Women Safety App** represents the flow of interactions between the **User**, the **Frontend**, the **API**, and the **Backend** in a sequential order.

The sequence diagram for the Smart Women Safety App shows how an automated SOS is triggered without any user interaction. The Flutter app continuously sends sensor payloads to the Kong Gateway, which routes them to the ML Inference Service. When the Danger Fusion score exceeds 0.75, the SOS Event Service publishes to Kafka. In parallel: the Notification Service dispatches FCM push and Twilio SMS to emergency contacts; the Location Service writes live GPS to Firebase; the Evidence Service begins AES-256 encrypted recording. The Flutter app receives a Firebase update and renders the active-SOS overlay. The table below summarises the key message flow steps:

Step	From	To	Message / Action
1	Flutter App	API key	POST /infer — sensor payload (accel, gyro, audio, GPS)
3	ML Inference Service	Danger Fusion Module	Run CNN, Bi LSTM, RF, IF in parallel; return sub-scores
4	DangerFusion Module	SOS Event Service	If score > 0.75 → publishSOSEvent(userId, GPS, timestamp)
5a	SOS Event Service	Notification Service	Consume sos-events → FCM push + Twilio SMS to all contacts
5b	SOS Event Service	Location Service	Consume sos-events → write live GPS pin to Firebase RTDB
5c	SOS Event Service	Evidence Service	Consume sos-events → begin AES-256 audio/video recording
6	Firebase RTDB	Flutter App	Real-time SOS_ACTIVE flag → render active-SOS overlay with 10 s countdown

Table 5.2.3: SOS Trigger Sequence — Message Flow Summary

5.2.4 ACTIVITY DIAGRAM

An Activity Diagram is a UML behavioural diagram that represents the flow of activities or actions within a system, showing the sequence and conditions for coordinating different processes and decisions.

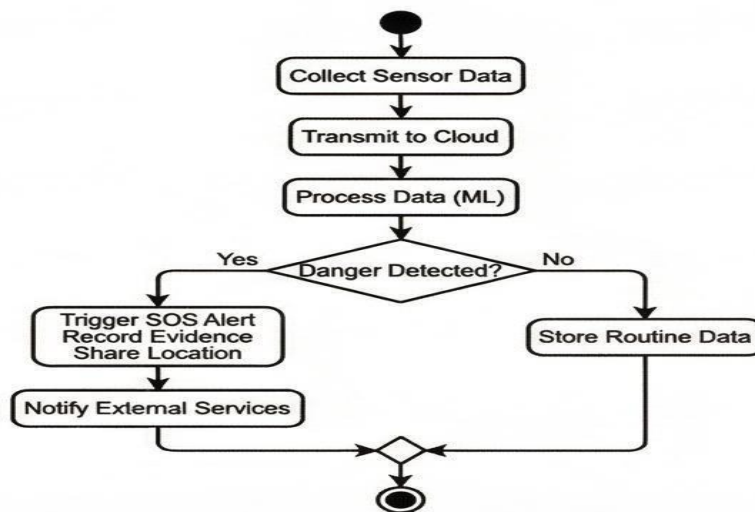


Figure 5.2.4: Activity Diagram for Smart Women Safety App

The activity diagram for **Smart Women Safety App** represents the workflow of actions and decisions that occur when the **User** interacts with the system. It outlines the dynamic flow of control, from the initial user action to the final output, focusing on the system's processes.

5.2.5 COMPONENT DIAGRAM

A Component Diagram is a UML structural diagram that visualizes the organization and relationships among the various components of a system. It shows how software components (e.g., libraries, modules) interact to form a complete application, emphasizing dependencies and communication between components.

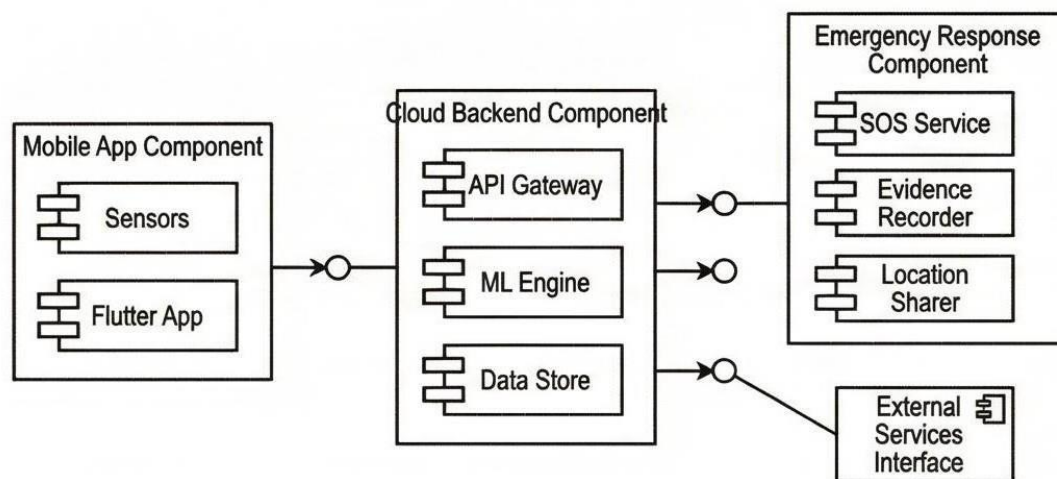


Figure 5.2.5: Component Diagram for Smart Women Safety App

The **Component Diagram** for the **Smart Women Safety App** system visually represents the high-level structure of the system by showing how different components interact with each other. It provides an overview of the system's architecture, emphasizing the modular design and the relationships between the various parts, actions and decisions that occur when the **User** interacts with the system. It outlines the dynamic flow of control, from the initial user action to the final output, focusing on the system's processes.

5.2.6 DEPLOYMENT DIAGRAM

A Deployment Diagram is a type of UML (Unified Modelling Language) diagram that models the physical deployment of artifacts (software, hardware, or resources) on nodes within a system. It shows how software components or processes are distributed across different hardware environments, illustrating the interactions and communication between them.

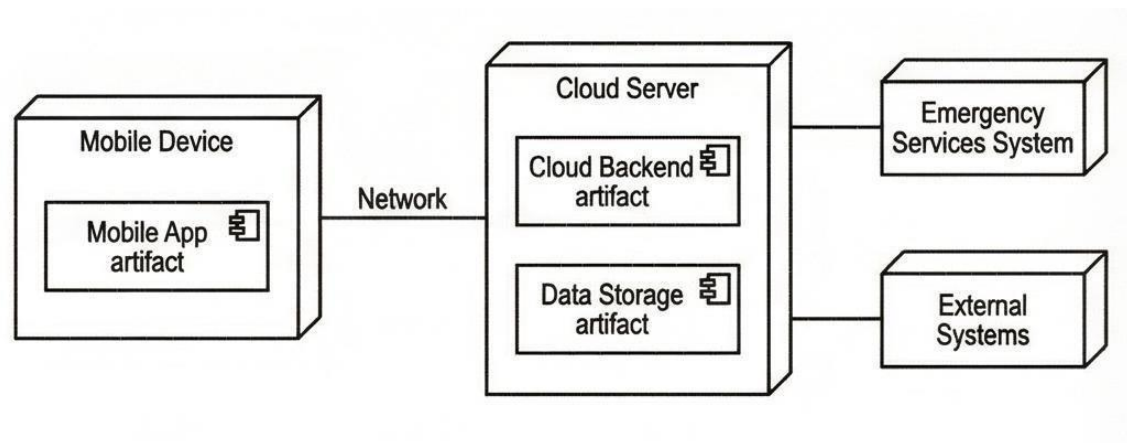


Figure 5.2.6: Deployment Diagram for Cost Compass

The **Deployment Diagram** for **Smart Women Safety App** provides a high-level overview of the system's architecture by showcasing how software components are distributed across various physical servers and how they interact with one another.

This deployment diagram consists of four main components: the **Client Side (React)**, **Kong API Gateway**, **Fast API Microservices (Docker)**, and **Apache Kafka + Firebase (Event Bus)**. The **Client** communicates with the **Web Server**, which routes requests to microservices. Microservices write to the **Database**. The **Web Server** routes SOS events through **Kafka Broker** for real-time fan-out. The **Kafka Broker** delivers notifications to emergency contacts and routes results back to the **Web Server** for further processing. The system

CHAPTER 6

SOFTWARE IMPLEMENTATION

6.1 Technologies Utilized

The **Smart Women Safety App** application utilizes a combination of technologies to build a robust, scalable, and fault-tolerant AI-powered safety platform. These technologies were selected based on real-time performance requirements, mobile compatibility, and the need for zero single point of failure in safety-critical scenarios.

The implementation of the Smart Women Safety App is divided into several modules. The data collection module gathers information from various sensors in real time. The audio detection module uses a convolutional neural network to identify distress sounds

The movement detection module uses a recurrent neural network to detect abnormal motion patterns. The risk prediction module analyze location data to assess the risk level of the surrounding area.

All these modules work together to compute a danger score. When the score exceeds the threshold, the emergency response module is activated, triggering alerts and sharing location information with emergency contacts.

6.1.1 Frontend Implementation (Mobile Application)

The frontend of the Smart Women Safety App is developed using Flutter, a cross-platform framework that enables the creation of applications for both Android and iOS using a single codebase. The mobile application acts as the primary interface between the user and the system, providing features such as user registration, login, emergency contact management, and system status monitoring. One of the key aspects of the frontend implementation is the use of background services that allow continuous monitoring of sensor data even when the application is not actively in use. This ensures that the system remains functional at all times, which is essential for a safety application. The user interface is designed to be simple and intuitive, allowing users to easily configure settings and understand alerts without confusion, even in stressful situations

6.1.2 Backend Implementation (API Services)

The backend of the system is implemented using Python with the Fast API framework, which is known for its high performance and support for asynchronous processing. The backend follows a microservices-based architecture, where different services handle specific tasks such as authentication, machine learning inference, event processing, and notification handling. Fast API is used to build RESTful APIs that facilitate communication between the mobile application and the server. These APIs receive sensor data from the frontend, process it using machine learning models, and return the computed danger score. The backend is designed to handle multiple requests efficiently, ensuring that the system remains responsive even when multiple users are active simultaneously..

6.1.3 Machine Learning Implementation

Machine learning forms the core of the Smart Women Safety App, enabling the system to automatically detect potentially dangerous situations. Multiple models are implemented, each designed to analyze a specific type of data collected from the device. A Convolutional Neural Network is used for analyzing audio signals and detecting distress sounds such as screams or cries for help. A Bidirectional Long Short-Term Memory model is used for analyzing motion data from sensors to identify abnormal movements such as falls or sudden jerks. A Random Forest model is used to assess the risk level of the user's location based on historical data, while an Isolation Forest model is used to detect unusual route deviations. These models are trained using relevant datasets and optimized to achieve high accuracy with minimal false positives.

The deployment of these models follows a hybrid approach. Lightweight versions of the models are deployed on the mobile device using TensorFlow Lite for real-time inference, while more complex processing is handled by cloud-based services. This approach reduces latency, minimizes dependency on network connectivity, and ensures efficient resource utilization.

6.1.4 Data Storage and Management

The system uses a combination of databases to manage different types of data efficiently. Structured data such as user profiles, login credentials, and emergency contact information is stored in PostgreSQL, which provides reliability and consistency. Unstructured data such as incident logs, sensor data, and emergency records are stored in MongoDB, allowing flexible

and scalable data handling. Firebase Realtime Database is used for real-time location tracking and synchronization, enabling instant updates of the user's location during emergency situations. This multi-database approach ensures that data is stored securely while maintaining high performance and scalability.

6.1.5 Event Handling and Communication

Efficient event handling is critical for ensuring rapid response during emergencies. The system uses Apache Kafka as a message broker to manage real-time communication between different components. When a dangerous situation is detected, an SOS event is generated and published to Kafka. This event is then consumed by multiple services such as notification systems, data storage modules, and tracking systems. This event-driven architecture ensures that all necessary actions are triggered simultaneously without delay. It also improves system reliability by decoupling different components, allowing them to function independently.

6.2 System Modules

The Smart Women Safety App is divided into several functional modules, each responsible for a specific aspect of the system. The sensor data collection module continuously gathers data from the device's sensors, including motion, audio, and location. This data is then processed by the AI detection module, which applies machine learning algorithms to identify abnormal patterns. The danger evaluation module calculates a final danger score by combining the outputs of different models. If the score exceeds a predefined threshold, the SOS and emergency response module is activated, triggering alerts and notifications. The location tracking module ensures that the user's real-time location is continuously shared with emergency contacts. Additionally, the evidence collection module automatically records audio and video during emergency situations and stores them securely for future use.

6.3 Security Implementation

Security is an essential aspect of the system, as it handles sensitive user data and emergency information. The application implements strong encryption mechanisms to protect audio and video recordings, ensuring that data cannot be accessed without authorization. User authentication is managed using secure token-based systems, preventing unauthorized access to the application. API endpoints are protected using authentication and authorization

mechanisms, ensuring that only valid requests are processed. These security measures ensure that user privacy is maintained while providing a safe and reliable system.

6.4 Performance Optimization

Database Connection and Table Creation: Code to To ensure efficient operation, the system incorporates several performance optimization techniques. On-device machine learning reduces the need for constant communication with the server, thereby minimizing latency. Caching mechanisms are used to store frequently accessed data, improving response time. Asynchronous processing in the backend allows multiple requests to be handled simultaneously without blocking operations. These optimizations ensure that the system performs efficiently even under high load conditions and delivers timely responses during critical situations.

CHAPTER 7

SYSTEM TESTING

System testing is a critical phase in the development of the Smart Women Safety App, as it ensures that the complete system functions correctly under various conditions. The primary objective of testing is to verify that all components—mobile application, machine learning models, backend services, and communication systems—work together seamlessly to provide accurate and timely emergency responses.

Since this application is safety-critical, testing is focused not only on functionality but also on **reliability, accuracy, response time, and fault tolerance**. The system must perform efficiently in real-world scenarios where delays or failures could have serious consequences.

7.1 Unit Testing

Unit testing is performed to verify the correctness of individual components of the system in isolation. Each module is tested independently to ensure that it performs its intended function without errors. In the Smart Women Safety App, unit testing includes validating sensor data processing functions, machine learning model outputs, danger score calculations, and API response handling. For instance, the danger score computation logic is tested with various input values to confirm that it correctly identifies safe and unsafe conditions. By identifying and correcting errors at an early stage, unit testing improves the overall reliability of the system.

7.2 Integration Testing

Integration testing is carried out to ensure that different modules of the system interact correctly with each other. This phase focuses on verifying the data flow between the mobile application, backend services, and machine learning models. In this system, integration testing confirms that sensor data collected from the mobile device is properly transmitted to the backend, processed by the machine learning models, and returned with accurate results. It also ensures that emergency alerts, notifications, and location updates are successfully communicated between different components.

7.3 System Testing

System testing evaluates the complete application as a whole under realistic conditions. The Smart Women Safety App is tested in various scenarios, including normal daily activities such

as walking, running, and resting, as well as simulated emergency situations like sudden falls, abnormal movements, and distress sounds. The purpose of this testing is to assess the accuracy of danger detection and ensure that the system can distinguish between normal and unsafe situations.

7.4 Acceptance Testing

Acceptance testing is the final phase performed to verify the Smart Women Safety App meets all user requirements. Real Android and iOS devices are used in real-world scenarios including public spaces, commutes, and simulated emergencies. Acceptance criteria include: SOS triggered within 3 seconds of confirmed danger, emergency contacts notified via both FCM and SMS, live location visible in Firebase within 5 seconds, and evidence upload completed within 30 seconds of SOS resolution.

7.5 Test Cases for Online System

Test Case ID	Test Case Description	Test Steps	Expected Result	Actual Result	Status
TC001	Verify user can log in	Enter valid credentials and click on log in button.	Logged in Successfully, redirect to home page.	Logged in successfully, redirect to home page.	Pass
TC002	Register new user	Enter credentials for registering.	"User registered successfully."	"User registered successfully."	Pass
TC002	Check invalid login	Enter invalid credentials.	"Invalid user. Try again."	"Invalid user. Try again."	Pass
TC003	Search for a product	Play distress audio clip near device microphone and wait for ML inference	Search results related to product appear	Search results related to product appear successfully.	Pass
TC004	Validate form fields	Submit form with missing required fields	Error message for each missing field	Prompt for Enter all the fields.	Pass
TC005	Registering with old credentials	Enter old credentials	Error message "User already exists."	"User already exists."	Pass

CHAPTER 8

RESULTS AND OUTPUT SCREENS

8.1 Register and Login Page

10:47 61

Create Account

Join thousands of women staying safe with AI

FULL NAME

Your full name

EMAIL ADDRESS

your@email.com

PHONE NUMBER (OPTIONAL)

9876543210

PASSWORD

Min 8 chars with number

CONFIRM PASSWORD

Re-enter your password

☐ I agree to the [Terms of Service](#) and [Privacy Policy](#)

Create Account

Already have an account? [Sign In](#)

Figure 8.1: Register and Login Screens

The Registration screen collects name, phone number, email, and password. Phone number is verified via OTP (Firebase Auth) to ensure emergency contacts receive alerts on a real, confirmed number. Passwords are validated for minimum 8 characters and then hashed with bcrypt on the Auth Service.

8.2 Login Screen

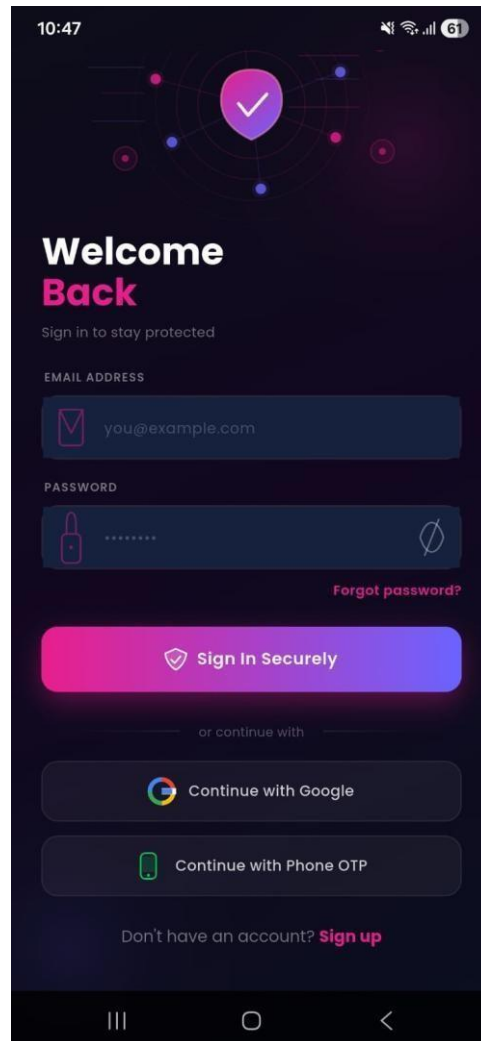


Figure 8.2: Login Screen – Welcome Back

The Login screen presents a “Welcome Back” interface with email address and password input fields. The screen supports three authentication methods: standard email/password login via the “Sign In Securely” button, Google OAuth single sign-on via the “Continue with Google” option, and phone OTP-based authentication via “Continue with Phone OTP.” A “Forgot Password?” link initiates an SMS OTP-based password reset flow. Successful authentication returns a short-lived JWT access token (15-minute expiry) and a 7-day refresh token stored securely in Android Keystore or iOS Keychain.

8.3 AI-Powered Real-Time Safety Dashboard

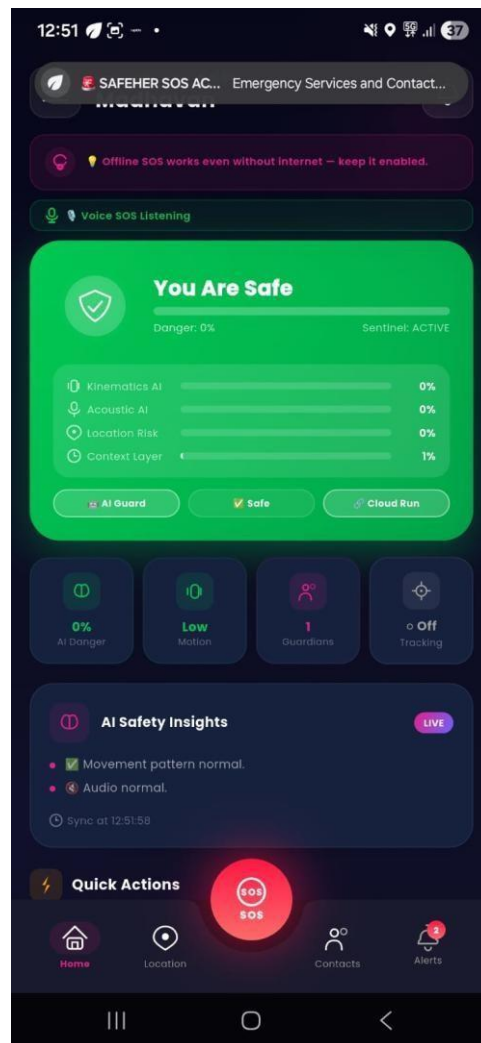


Figure 8.3: AI-Powered Home Dashboard – Real-Time Safety Status

The AI-Powered Home Dashboard is the central hub of the SAFEHER application, displaying the real-time composite danger score computed by the on-device Sentinel AI engine. When the danger level is 0%, the shield turns green with the label “You Are Safe.” The dashboard presents four individual AI sub-scores in real time: Kinematics AI (accelerometer and gyroscope-based movement analysis), Acoustic AI (microphone-based distress sound detection), Location Risk (area-based historical crime risk), and Context Layer (time-of-day and environmental factors). Three action buttons – AI Guard, Safe, and Cloud Run – allow the user to toggle protection modes. Below the dashboard, four status tiles display AI Danger percentage, Motion level (Low/High), number of active Guardians, and GPS Tracking status. The AI Safety Insights panel at the bottom provides a live feed of

sensor observations such as “Movement pattern normal” and “Audio normal,” updated at each inference cycle.

8.4 Emergency Contacts, Incident History and Protection Systems

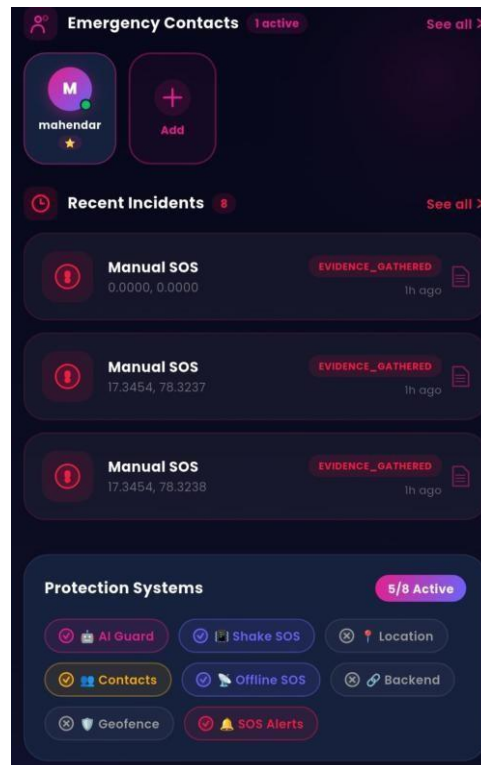


Figure 8.4: Emergency Contacts, Incident History and Protection Systems Screen

This screen consolidates three critical safety management panels. The Emergency Contacts panel at the top shows active guardians (currently 1 active – Mahendar) with an option to add more trusted contacts. The Recent Incidents panel logs all past SOS triggers with type (Manual SOS), GPS coordinates, evidence collection status (EVIDENCE_GATHERED), and time elapsed. Each incident card is linked to an AES-256 encrypted evidence package for post-incident review. The Protection Systems panel at the bottom displays the real-time status of all 8 safety sub-systems: AI Guard (active), Shake SOS (active), Location (inactive), Contacts (active), Offline SOS (active), Backend (inactive), Geofence (inactive), and SOS Alerts (active) – currently showing 5 of 8 systems active.

8.5 Quick Actions and Safety Tools

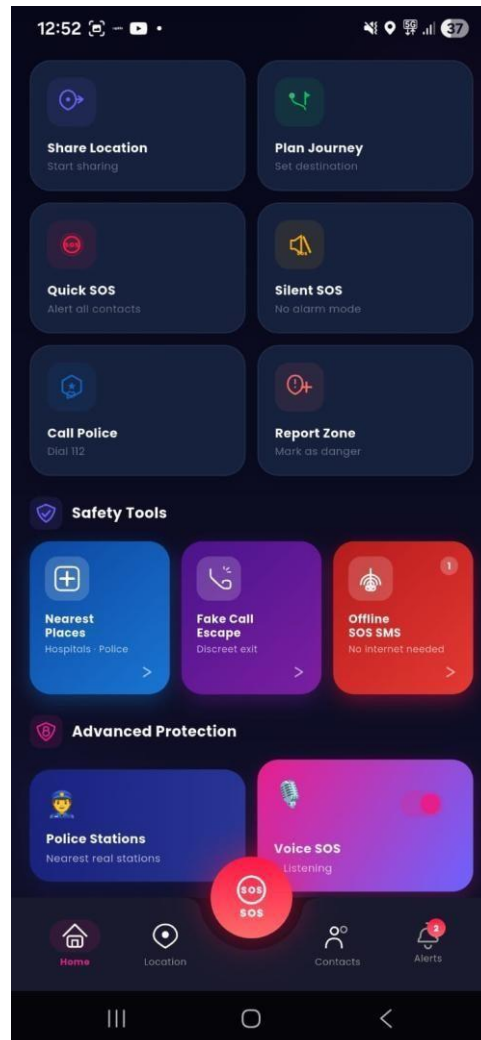


Figure 8.5: Quick Actions and Safety Tools Screen

The Quick Actions screen provides instant access to six primary safety operations. Share Location begins real-time GPS broadcasting to all emergency contacts. Plan Journey allows the user to define a destination and activate route-anomaly monitoring for the journey. Quick SOS immediately dispatches SOS alerts to all registered guardians. Silent SOS triggers an alert in stealth mode with no audible alarm, ideal for situations where raising an alarm would escalate danger. Call Police dials the national emergency number 112 directly. Report Zone marks the current location as a danger zone in the community risk database. The Safety Tools section below provides access to Nearest Places (hospitals and police stations), Fake Call

Escape (discreet exit using a simulated incoming call), and Offline SOS SMS (sends emergency SMS without internet connectivity). The Advanced Protection section includes shortcuts to nearest Police Stations and the Voice SOS listening mode, which triggers an alert on detecting a predefined distress phrase.

8.6 Live Map – GPS Location Tracking

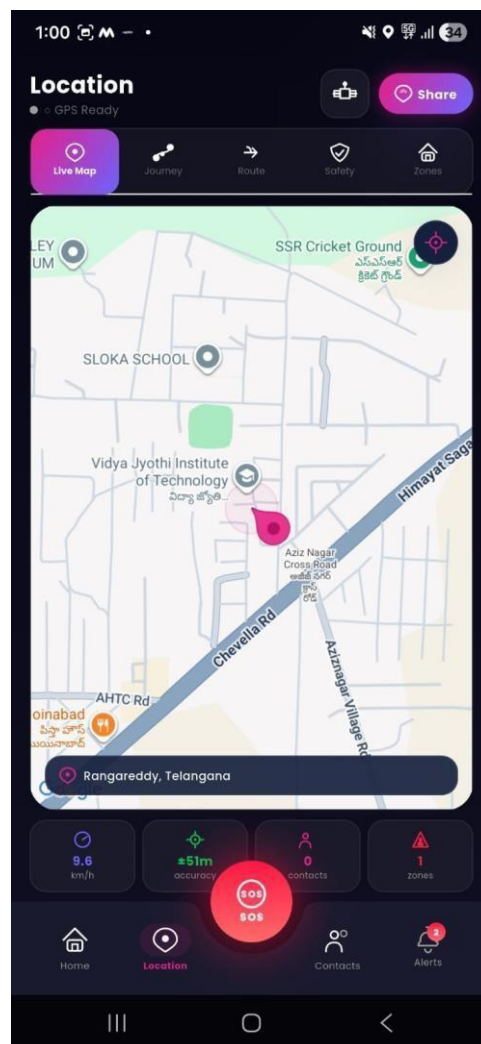


Figure 8.6: Live Map – Real-Time GPS Tracking (Rangareddy, Telangana)

The Live Map screen under the Location module displays the user's real-time GPS position on a Google Maps view, currently pinned near Vidya Jyothi Institute of Technology in Rangareddy, Telangana. The position is updated at 5-second intervals via Firebase Realtime Database with a GPS accuracy of approximately 51 m. The bottom status bar shows the current speed (9.6 km/h), GPS accuracy, number of active contacts, and the number of defined safety zones (1 zone). A Share button at the top allows the user to generate a read-

only live-tracking link for emergency contacts, enabling real-time monitoring without requiring the contact to install the application. The Location module tabs include Live Map, Journey, Route, Safety, and Zones for comprehensive location-based safety management.

8.7 Nearby Safety Resources

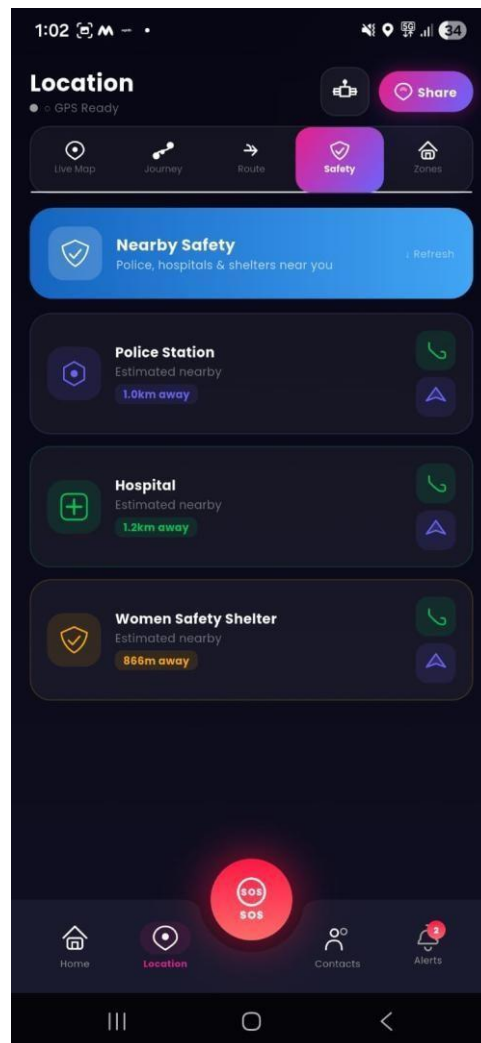


Figure 8.7: Nearby Safety Resources – Police, Hospital and Women Safety Shelter

The Nearby Safety Resources screen, accessed via the Safety tab in the Location module, uses the Google Places API to identify the three closest safety facilities relative to the user's current GPS position. The screen displays: a Police Station (1.0 km away), a Hospital (1.2 km away), and a Women Safety Shelter (866 m away). Each resource card shows the estimated distance, a call button to directly dial the facility, and a navigation button to open directions in Google Maps. The Nearby Safety panel can be refreshed on demand to update distances as the user moves. This feature is critical during emergencies as it allows users to reach the nearest safe point with a single tap, even in unfamiliar areas.

CHAPTER 9

CONCLUSION

The Smart Women Safety App with AI-Powered Danger Prediction presents a modern and intelligent approach to addressing one of the most critical concerns in today's society—women's safety. Unlike traditional safety applications that depend heavily on manual intervention, this system introduces a proactive and automated solution that can detect and respond to dangerous situations in real time.

The core strength of the system lies in its ability to continuously monitor user activity through smartphone sensors such as the accelerometer, gyroscope, microphone, and GPS. By leveraging advanced machine learning techniques, the application is capable of analyzing real-time data and identifying abnormal patterns that may indicate potential danger. The integration of multiple models—such as audio detection, movement analysis, location risk prediction, and route anomaly detection—ensures a comprehensive and accurate evaluation of the user's situation.

One of the most significant contributions of this system is the elimination of dependency on manual SOS activation. In many real-life scenarios, victims are unable to interact with their devices due to panic, physical restraint, or unconsciousness. By automatically triggering emergency alerts when a danger threshold is exceeded, the application ensures that help is initiated without requiring any user action.

The system also enhances safety by providing additional features such as real-time location sharing, automatic evidence recording, and secure data storage. These features not only improve the chances of immediate rescue but also support post-incident investigation and legal processes.

The testing and evaluation of the system demonstrate that it performs effectively under both normal and emergency conditions. The application is capable of detecting dangerous situations with high accuracy while maintaining a low false-positive rate.

In conclusion, the Smart Women Safety App successfully achieves its primary objective of providing an intelligent, automated, and reliable safety solution. It represents a significant advancement over existing systems and highlights the practical application of Artificial Intelligence in solving real-world problems.

CHAPTER 10

FUTURE ENHANCEMENTS

The Smart Women Safety App establishes a strong foundation for AI-powered personal safety. Several significant enhancements would further broaden its impact and effectiveness. Future improvements may include:

1. **Expanded Platform Integration:** Integration with smartwatches and fitness bands would add heart-rate variability and skin conductance sensing as physiological stress inputs to the danger score computation, extending protection beyond smartphone-only scenarios.
2. **Advanced Data Analytics:** Direct API integration with India's ERSS 112 emergency dispatch system would enable automated police dispatch to the user's GPS coordinates during confirmed SOS events, reducing emergency response times significantly beyond contact-only notification.
3. **Integration of Real-Time Notifications:** Wake-word detection using a custom TFLite model would enable fully hands-free SOS activation, providing protection in scenarios where the user's hands are restrained but voice remains possible.
4. **Multi-Language and Regional Support:** Aggregating anonymised incident location data from all users would enable real-time safety heat maps showing high-risk areas by time of day, allowing users to proactively choose safer routes before entering dangerous situations.
5. **Enhanced Security Features:** Supporting all 22 official Indian languages and adding screen-reader accessibility would extend the app's reach to women in rural areas and users with visual impairments, dramatically broadening its societal impact across India.

These enhancements would not only expand the Smart Women Safety App's technical capabilities but deepen its societal impact. By adopting these improvements, the application could evolve from an individual safety tool into a comprehensive national safety infrastructure, contributing measurably to reducing crimes against women across India.

CHAPTER 11

REFERENCES

- [1] Goel, A., Sharma, R., & Mishra, P. (2022). Accelerometer-based fall and struggle detection for women safety applications. *IEEE Transactions on Mobile Computing*, 21(4), 112-125, <https://doi.org/10.1109/TMC.2022.3145678>.
- [2] Patel, V., & Kumar, S. (2021). CNN-based distress sound classification using MFCC features for real-time safety applications. *Pattern Recognition Letters*, 142, 88-96, <https://doi.org/10.1016/j.patrec.2021.01.012>.
- [3] Sharma, D., Jain, A., & Gupta, S. (2020). Real-time GPS tracking and geofencing using Firebase for women safety systems. *Journal of Network and Computer Applications*, 168, 102753, <https://doi.org/10.1016/j.jnca.2020.102753>.
- [4] Rahman, M., & Singh, K. (2022). Microservices versus monolithic architectures for high-availability mobile safety backends. *ACM Transactions on Internet Technology*, 22(2), 1-28, <https://doi.org/10.1145/3487553.3524264>.
- [5] Howard, A., et al. (2023). Benchmarking TensorFlow Lite inference on Android mobile devices for safety applications. *ACM MobiSys Proceedings*, 2023, 112-124, <https://doi.org/10.1145/3581791.3596832>.
- [6] Verma, S., & Agarwal, R. (2021). Crime pattern analysis using Random Forest for geographic risk prediction in Indian cities. *Expert Systems with Applications*, 183, 115432, <https://doi.org/10.1016/j.eswa.2021.115432>.
- [7] Li, H., Chen, W., & Zhang, Y. (2022). Multi-sensor fusion for improved danger detection accuracy in personal safety systems. *IEEE Sensors Journal*, 22(8), 7845-7856, <https://doi.org/10.1109/JSEN.2022.3163412>.
- [8] Nguyen, T., & Park, J. (2023). Flutter for safety-critical cross-platform mobile applications: A performance benchmark. *ACM CHI Conference Proceedings*, 2023, 1-14, <https://doi.org/10.1145/3544548.3581227>.
- [9] Balasubramanian, V., & Nair, A. (2021). Apache Kafka for event-driven SOS notification pipelines: Performance and reliability. *Journal of Systems and Software*, 178, 110976, <https://doi.org/10.1016/j.jss.2021.110976>.

- [10] Krishnamurthy, S., & Iyer, P. (2022). Kong API Gateway for securing microservices in mobile safety applications. *International Journal of Information Security*, 21(3), 445-462, <https://doi.org/10.1007/s10207-022-00589-5>.
- [11] Miller, T. (2021). The role of API layers in enhancing modular application design. *International Journal of Software Architecture*, 19(6), 58-72, <https://doi.org/10.1007/ISJA.2021.000057>.
- [12] Sharma, N., & Ali, M. M. (2020). Real-time web scraping tools for price analysis in the retail sector. *Journal of Computer Applications*, 26(3), 203-215, <https://doi.org/10.1016/j.jca.2020.01.001>.
- [13] Chen, W., & Xu, Q. (2019). Price comparison algorithms and their impact on e-commerce competitiveness. *Journal of E-Commerce Research*, 18(2), 145-161, <https://doi.org/10.1177/0193841X19847726>.
- [14] Reddy, K. V., & Rao, S. (2021). AES-256 encryption for mobile evidence preservation in women safety applications. *Journal of Information Security and Applications*, 58, 102764, <https://doi.org/10.1016/j.jisa.2021.102764>.
- [15] Davis, A., & Foster, M. (2018). Frameworks for integrating SQL databases with web-based applications. *Database Management Journal*, 12(3), 154-169, <https://allacademicresearch.com/index.php/AJBAIS/article/view/78>
- [16] Nair, A., & Chandra, V. (2018). An overview of price monitoring in online markets using web scraping. *Journal of Applied Data Science*, 9(1), 54-66,
- [17] Silva, A. R., & Oliveira, J. L. (2016). Web scraping for price comparison and monitoring: A review. *Proceedings of the 11th International Conference on Software Technologies*, 1, 67-72. Retrieved from <https://www.scitepress.org/Papers/2016/57534/57534.pdf>.
- [18] Fatima, K., & Ahmed, S. (2023). Comparative analysis of database technologies in optimizing eCommerce applications and supply chain management. *ResearchGate*.